Scope of This Research — Table Test

Table 1: Scope boundary for the full research programme and its relation to future work.

Research Activity	In Scope	Out of Scope	How This Study Enables Future Work
Core Reactor Characterisation			
Industry requirements matrix	Yes	—	Defines power, range, and energy density targets that bound all downstream reactor design
Reaction parameter characterisation (T, P, particle size, fuel-to-water ratio)	Yes (Tier 1)	—	Produces the experimental dataset for reactor geometry and operating point selection
By-product composition analysis ($Al(OH)_3$, $AlOOH$, Al_2O_3)	Yes (Tier 1)	—	Determines achievable effective energy density and recycling compatibility under real operating conditions
Seawater vs. distilled water reactant comparison	Yes (Tier 2)	—	Establishes whether onboard seawater can be used directly, and specifies any pre-treatment or filtration requirements
Reactor start-up time and transient load characterisation	Yes (Tier 1)	—	Provides the open-loop dynamic data required for control system design and battery buffer specification
Marine safety protocols (bunkering, dust control, emergency response)	Yes	—	Provides the safety evidential basis for future certification submissions to RINA, LR, and DNV
Reactor geometry and material selection	Yes (lab-scale, indicative)	Full-scale design	Lab-scale findings provide the basis for scaled engineering design in future work
Preliminary cost estimation (reactor and aluminium fuel)	Yes (indicative, order-of-magnitude)	Final production costs	High-level estimates inform the economic feasibility case and identify where further cost refinement is needed
Control Systems			
Closed-loop control system development	Yes (fixed-batch demonstrator)	Production-ready control hardware	Demonstrates thermal and kinetic controllability; quantifies response lags, rise times, and stability boundaries

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Research Activity	**In Scope**	**Out of Scope**	**How This Study Enables Future Work**
Microcontroller hardware selection and implementation	Yes	Maritime ruggedisation	Provides a validated testbed for control algorithm development; hardware choices inform future procurement specifications
Power mode transition characterisation	Yes	Optimal controller design	Quantifies the reactor's dynamic response to step changes, enabling future engineers to size battery buffers and design control systems
Mode granularity optimisation	Yes	—	Determines the optimal number and spacing of discrete power modes for marine propulsion, balancing controllability and manoeuvrability
Engine and Battery Analysis			
Thermal integration pathway trade-off (direct vs. separation)	Yes	Detailed separation system design	Identifies the lowest-risk pathway for converting the steam-H_2 mixture into shaft power
Prime mover survey and selection	Yes (requirements definition)	Engine design or integration	Produces a decision matrix comparing mature, certifiable prime movers (steam turbine, ORC, diesel)
Battery buffer specification	Yes (requirements definition)	Battery system design	Defines minimum energy (kWh) and power (kW) requirements for a marine-certified battery to bridge reactor transients
Hybrid architecture recommendation	Yes (requirements definition)	Hybrid system integration	Recommends the simplest hybrid topology (series, parallel, or combined) that meets buffering requirements with lowest system risk
Efficiency target relaxation analysis	Yes	Efficiency optimisation	Establishes realistic, defensible efficiency targets (20–30%) for a Generation-1 prototype
Out of Scope (Future Work)			
Full-scale reactor design and stress testing	—	Yes	Lab-scale findings from this study provide the basis for scaling engineering design in future work
Continuous-flow reactor (real-time fuel injection & solids handling)	—	Yes	Fixed-batch control data demonstrates fundamental controllability; continuous-flow engineering is a separate mechanical design challenge
Engine and turbine detailed design and integration	—	Yes	Interface specifications (steam temperature, pressure, flow rate) defined here enable future procurement and integration

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Research Activity	**In Scope**	**Out of Scope**	**How This Study Enables Future Work**
Battery system detailed design and BMS development	—	Yes	Capacity and power requirements defined here enable future procurement and integration
Classification society approval (full-scale)	—	Yes	Safety protocols and experimental datasets developed here form the evidential basis for future certification submissions

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